Sample Knowledge Base for RAG Testing

Introduction to Python in AI

Python is a high-level, interpreted programming language known for its simplicity and readability. It has become the undisputed lingua franca for Artificial Intelligence (AI) and Machine Learning (ML) development due to its rich ecosystem of libraries such as NumPy, Pandas, Scikit-learn, and TensorFlow.

The Role of FastAPI

FastAPI is a modern, high-performance web framework for building APIs with Python. It is based on standard Python type hints and automatically generates interactive API documentation (Swagger UI). In AI application engineering, FastAPI is widely used to serve model predictions and RAG endpoints reliably and swiftly.

Understanding Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a powerful architectural pattern designed to improve the accuracy and reliability of Large Language Models (LLMs). Instead of relying solely on the model's internal parametric memory—which can lead to hallucinations—a RAG system dynamically retrieves relevant context from a private knowledge base (like a Vector Database) and provides it alongside the user's prompt to generate grounded, factual responses.

Vector Databases and ChromaDB

Vector databases store high-dimensional numeric arrays called embeddings. ChromaDB is an open-source, lightweight vector database designed specifically to integrate seamlessly with AI applications. It allows developers to store text chunks along with their corresponding vector embeddings and perform lightning-fast semantic similarity searches using Cosine Similarity.